

Huy Xuan Hoang Nguyen

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Education

UNIVERSITY OF SOUTHERN CALIFORNIA

Expected May 2026

- *Master of Science* in Computer Science
- *Related Courseworks:* Analysis of Algorithms, Foundation of Artificial Intelligence, 3D Graphic Rendering

UNIVERSITY OF CALIFORNIA, DAVIS

September 2020 - June 2024

- *Bachelor of Science* in Computer Science
- *Related Courseworks:* Data Structures (C & C++), Object-Oriented Programming, Artificial Intelligence, Machine Learning (Python), Operating System, Gameplay Programming (Unity), Foundations of Algorithms

Skills

- **Proficient Programming Languages:** Python, C/C++, C#, Bash, JavaScript, SQL
- **Database Management:** SQL query optimization, PostgreSQL configuration, database security policies
- **Systems & Framework:** Arduino IDE, AutoCAD, RTMaps, Git, Docker, Terraform, TensorFlow, PyTorch
- **Cloud & Security:** AWS (S3, RDS, Rekognition, VPC), IAM Security Policies, TCP/IP fundamentals
- **Data & Research:** Anomaly detection (AWS), Data visualization (Matplotlib), Data Integration & Analysis
- **Certificate:** [AWS Certified Cloud Practitioner](#), [CodePath Intermediate Web Development Certificate](#)
- **DevOps & Collaboration:** Git/Github, Github Action (CI/CD), Jira, Confluence, Agile Workflow

Related Project

AUTOMATED VPC PROVISIONING FOR CLOUD DATABASE ACCESS

July 2025

AWS Cloud Security & Infrastructure - AWS - 30 hours

- Designed a Virtual Private Cloud infrastructure automation in AWS using Terraform and Boto3, including subnets, security groups, and internet gateways for a relational database application
- Engineered PostgreSQL database architectures including schema validation, analyzed access control policies, supporting scalable data infrastructure and reproducible deployment workflows
- Built a parameterized and reusable infrastructure system to support rapid provisioning across development, testing, and production environments, enabling scalable data infrastructure

AWS REKOGNITION IMAGE ANALYZER PLATFORM

July 2025

Cloud Security & AI Project - AWS - 20 hours

- Built image analysis application using Python, AWS S3, and AWS Rekognition to detect and classify faces and text, with potential applications in identity verification and access monitoring
- Utilized Matplotlib and Pillow to generate and overlay bounding boxes on detected objects, improving image analysis visualization for end users
- Implemented CI/CD pipelines with GitHub Actions to automate test execution, regression testing, and deployment, ensuring consistent build quality across releases

ORDER UP

September 2025 - December 2025

Lead Director & Game Designer - USC Game Design Project Pitch - 40 hours/week

- Led a 6-person team to develop a puzzle-based cooking simulation game in Unity with C#, featuring a unique recipe discovery mechanic where players deduce ingredients from dish names and images rather than following explicit recipes
- Architected core gameplay systems including drag-and-drop ingredient handling, multi-state cookware mechanics (oven, fryer, pot, pan), and a star-based rating system that scores dishes based on ingredient accuracy
- Built custom unit testing scripts and debug tools in C# to validate dishes spawning, targeting logic, and UI behavior across 3+ game states, improving feature verification efficiency

Experience

ECOCAR TEAM

January 2023 – June 2024

Software Developer - UC Davis EcoCAR - 40 hours/week

- Contributed to applied research and development of vehicle software modules in a national automotive competition sponsored by the U.S. Department of Energy and General Motors, collaborating with a multidisciplinary team to promote sustainability and innovation in vehicle systems
- Conducted software validation research on hardware/software integration for computer vision rendering related to communication failures and system instability
- Developed automated test and simulation tools in Python and C++ to perform functional & regression testing on vehicle perception systems for large-scale datasets, enabling early detection of anomalous behaviors